

In satellite meteorology and radiative transfer models like SARTA, **Jacobians** and **Weighting Functions** are essentially the "sensitivity blueprints" of a satellite sensor. [1, 2, 3]

When SARTA runs, it outputs a matrix that tells an inverse retrieval model exactly how to map the raw satellite data back into a vertical weather profile. [4]

What are Jacobian Weights mathematically?

The Jacobian matrix ($\mathbf{K}$) represents a massive grid of partial derivatives. It explicitly links **what the satellite sees** (changes in Brightness Temperature, ∂T_b) to **what is changing in the atmosphere** (changes in a specific layer's property, ∂X_L): [2, 4]

$$\mathbf{K} = \frac{\partial T_b}{\partial X_L}$$

- **Each Row** represents one specific satellite channel.
- **Each Column** represents one of the 100 atmospheric layers. [2]

If a number in that matrix is zero, it means that specific satellite channel is completely blind to that layer of the atmosphere. If the number is large, it means the channel is highly sensitive to that layer.

		COLUMNS (Atmospheric Layers: 1 to 100)				
		Layer 1 (Top) → → → → Layer 100 (Surface)				
ROWS to mid-atmosphere (Channels)	Channel 1	0.00	0.01	0.75	0.00	← High sensitivity
	Channel 2378	0.00	0.00	0.00	0.91	← Windows channel: sees the surface

The Two Pillars: Temperature vs. Gas Jacobians

1. Temperature Jacobians ($\mathbf{K}_T$) — The Traditional "Weighting Function"

Temperature Jacobians are shaped directly by the derivative of the layer transmittance ($\frac{dt}{d\ln p}$). They look like smooth, bell-shaped curves when plotted vertically: [2, 5, 6]

- **The Peak:** The peak of the bell curve tells you exactly where a channel's energy originates. For example, a core $15\ \mu\text{m}$ CO_2 channel might peak at $10\ \text{hPa}$ (the stratosphere), while a window channel peaks at the surface. [5, 6]

- **The Weight:** If a data assimilation system notices an unexpected ΔK jump in a stratospheric channel, it multiplies that jump by the K_T weights to know exactly which layers in the stratosphere actually warmed up.

2. Water Vapor (K_q) & Ozone (K_{O_3}) Jacobians

Gas Jacobians are more dynamic because the gas concentration itself is highly variable. They track how an increase or decrease in gas molecules shapes the overall signal: [7, 8]

- **The "Muting" Effect:** If you add more water vapor to a layer, it makes the layer more opaque. This blocks the warm thermal energy coming from the Earth's surface below. [6]
- **Negative Weights:** Because adding absorbing gas blocks warm upwelling light and replaces it with cold emission from higher up, gas Jacobians often output **negative values** in the troposphere. A negative weight tells the inversion engine: *"If this channel suddenly gets colder, it means the atmosphere is getting wetter, not necessarily colder."*

Why SARTA's "Analytical" Approach Matters

In older or more brute-force radiative transfer engines, calculating these weights required the **Perturbation Method**: [1]

1. Run the forward model once to get a baseline radiance.
2. Add a tiny bit of heat to Layer 1, run the model again, and calculate the difference.
3. Repeat that process for all 100 layers, across thousands of channels. [1]

This required performing tens of thousands of model runs per profile, making real-time weather forecasting mathematically impossible. [1]

Because SARTA's optical depths are based entirely on simple polynomial strings, it differentiates those strings mathematically on the fly. **SARTA outputs the full forward radiance and the entire Jacobian sensitivity matrix in a single step**, calculating the analytical calculus weights in mere fractions of a millisecond. [1]

Would you like to examine how these output Jacobians are mathematically combined with an **un-apodized instrument framework**, or should we look at how a weather forecast model uses them in **3D/4D-Var data assimilation**?

[1] <https://harvest.usask.ca>

[2] <https://cimss.ssec.wisc.edu>

[3] <https://docs.pytorch.org>

[4] <https://medium.com>

[5] <https://rammb2.cira.colostate.edu>

[6] <https://www.researchgate.net>

[7] <https://link.springer.com>

[8] <https://elib.dlr.de>